RIYA PATKAR

AI/ML & Full-Stack Developer

📧 riyapatkar20@gmail.com 📞 +91 7021475449 💻 github.com/riyax11 🌐 portfolio-riya-patkar.vercel.app

PROFESSIONAL SUMMARY

Final-year AI & Machine Learning engineering student who builds end-to-end products: NLP classifiers, LLM-powered analytics tools, computer-vision prototypes, and full-stack web applications. Seeking an AI/ML or Software Engineering internship to ship practical, user-focused features.

EDUCATION & TECHNICAL SKILLS

B.E., Artificial Intelligence and Machine Learning — Lokmanya Tilak College of Engineering, Navi Mumbai

Final Year (2026–27) | **CGPA: 7.80**

Languages:	Python, JavaScript, TypeScript, SQL
AI & ML:	Machine Learning, Deep Learning, NLP, Computer Vision, scikit-learn, YOLOv8, OpenCV, LLM Apps (Groq API)
Web & Backend:	React, Next.js, Node.js, Express, FastAPI, Django, REST APIs
Data & Tools:	PostgreSQL, SQLite, SQLAlchemy, Chart.js, Git, GitHub

FEATURED PROJECTS

VeriJob AI: Fake Job Posting Detector

`github.com/riyax11/verijob-ai`

Python, scikit-learn, NLTK, FastAPI, React

- **NLP Classification Engine:** Built an NLP classifier that flags fraudulent job postings using TF-IDF feature extraction and class-weighted training on heavily imbalanced data (~5% fake postings).
- **Model Optimization & Deployment:** Evaluated Logistic Regression, Random Forest, and Naive Bayes; deployed Logistic Regression achieving **91% recall** on fake postings, served via a FastAPI backend and interactive React interface.

FinSight AI: Natural-Language Banking Analytics

`github.com/riyax11/finsight-ai`

Django, PostgreSQL, Groq LLM, Chart.js

- **LLM-to-SQL Pipeline:** Built a natural language analytics platform where an LLM converts plain-English queries directly into optimized SQL executed against PostgreSQL with over 1 million records.
- **Interactive Insights & Safety:** Integrated an SQL query safety validation layer, dynamic Chart.js data visualizations, and AI-generated textual summaries while displaying generated SQL queries for transparent auditability.

ChainGuard: Blockchain Anti-Money Laundering System

`github.com/riyax11/chainguard-aml`

React, Node.js, Solidity, Hardhat

- **Smart Contract Risk Engine:** Developed a 5-rule transaction risk engine scoring transactions from 0 to 100, permanently writing risk scores to Solidity smart contracts as an immutable audit trail with role-based access control.
- **Full-Featured Dashboard:** Constructed a 6-page React compliance dashboard featuring real-time transaction alerts, SAR PDF export capabilities, a targeted watchlist, and dynamic fund-flow network graphs.

PROJECTS (CONTINUED)

DataForge AI: Governed Data Pipeline Workspace `github.com/riyax11/dataforge-ai`

React, Express, PostgreSQL, Drizzle, Groq

- **Governed Data Workflows:** Engineered a dataset workspace where incoming CSVs are profiled and converted into risk-classified execution plans; critical steps (deduplication, publishing) require explicit human approval logged in an immutable trail.
- **Privacy-Preserving LLM Planner:** Integrated a Groq LLM planner designed to operate strictly on column metadata (zero row data exposure), enforced strict approval guardrails, and built an automated fallback to rule-based planning upon API errors.

InventoryQR: Warehouse Inventory Tracker `github.com/riyax11/InventoryQR`

React, FastAPI, SQLAlchemy, SQLite

- **Barcode Tracking Infrastructure:** Developed a QR and barcode verification system that validates physical product locations against rack designations and accurately tracks inbound/outbound movements with detailed audit histories.

Chair Occupancy Detection Python, YOLOv8, OpenCV `github.com/riyax11/chair-occupancy-detection`

- **Computer Vision Occupancy Tracker:** Implemented a object-tracking algorithm utilizing pretrained YOLOv8 models to detect human presence and compute continuous occupancy duration for designated seating zones across public video feeds.

WORK EXPERIENCE

Reliance Jio — Video Analytics Intern (Business Automation) Jan 1, 2026 – Jan 31, 2026

- **Computer Vision Prototyping:** Engineered an automated chair occupancy tracking prototype using pretrained YOLOv8 models and OpenCV to continuously detect human presence and compute exact occupancy durations across defined chair zones.
- **Testing & Technical Review:** Validated detection accuracy across multiple public video camera angles, benchmarked tracking stability, and presented findings and architectural recommendations in the final internship report.

CERTIFICATIONS & TRAINING

Agentic AI: From Learner to Builder
IBM SkillsBuild with CSRBOX — 4-Week Project-Based Program | *July – August 2025*

Green Skills and Artificial Intelligence
Edunet Foundation, Skills4Future Program | *March 2025*